

SOLVERS MATHEMATICS — CLASS 7

CHAPTER 7: A TALE OF THREE INTERSECTING LINES (TRIANGLES)

CHAPTER ASSESSMENT TEST PAPER

Time Allowed: 1 Hour | **Maximum Marks:** 40 Marks

SECTION A: OBJECTIVE TYPE QUESTIONS (10 Marks)

I. Multiple Choice Questions (1 Mark each)

1. The sum of the interior angles of any triangle is always:

[1 Mark]

- (a) 90°
- (b) 180°
- (c) 270°
- (d) 360°

2. An exterior angle of a triangle is equal to:

[1 Mark]

- (a) Sum of its interior opposite angles
- (b) Sum of all interior angles
- (c) Difference of its interior opposite angles
- (d) 180° minus opposite angle

3. Which of the following sets of side lengths can form a valid triangle?

[1 Mark]

- (a) 2 cm, 3 cm, 5 cm
- (b) 3 cm, 6 cm, 10 cm
- (c) 6 cm, 8 cm, 10 cm
- (d) 4 cm, 4 cm, 9 cm

4. A triangle in which all three sides are of unequal lengths is known as:

[1 Mark]

- (a) Equilateral triangle
- (b) Isosceles triangle
- (c) Scalene triangle
- (d) Right-angled triangle

5. In a right-angled triangle, if one acute angle is 42° , the other acute angle is:

[1 Mark]

- (a) 42°
- (b) 48°
- (c) 138°
- (d) 90°

II. Fill in the Blanks (1 Mark each)

6. The line segment joining a vertex of a triangle to the midpoint of its opposite side is called a _____.

7. A triangle cannot have more than _____ obtuse angle.

[1 Mark]

[1 Mark]

8. In any triangle, the side opposite to the largest angle is always the _____ side.

III. True or False (1 Mark each)

[1 Mark]

9. The sum of any two sides of a triangle must be strictly greater than the third side.

[1 Mark]

10. An equilateral triangle has each angle measuring 60° .

[1 Mark]

SECTION B: VERY SHORT ANSWER QUESTIONS (6 Marks)

11. The angles of a triangle are in the ratio $1 : 2 : 3$. Find the measure of all three angles.

[2 Mark]

12. In $\triangle ABC$, the exterior angle at vertex C measures 110° . If one interior opposite angle is 45° , find the other interior opposite angle.

[2 Mark]

13. Can a triangle have angles measuring 65° , 75° , and 50° ? Justify your answer.

[2 Mark]

SECTION C: SHORT ANSWER QUESTIONS (12 Marks)

14. One of the acute angles of a right-angled triangle is twice the other. Find the measures of both acute angles.

15. In $\triangle PQR$, the bisector of $\angle P$ meets QR at S . If $\angle Q = 70^\circ$ and $\angle R = 30^\circ$, find $\angle PSR$ and $\angle PSQ$.

[3 Mark]

16. The lengths of two sides of a triangle are 8 cm and 11 cm . Between which two numbers must the length of the third side fall?

[3 Mark]

[3 Mark]

17. In the given figure, find the values of the unknown variables x and y :

Exterior angle is 120° , one interior opposite angle is x , the adjacent interior angle is y , and the second interior opposite angle is 50° .

[3 Mark]

SECTION D: LONG ANSWER QUESTIONS (8 Marks)

18. (a) State and prove the **Angle Sum Property of a Triangle** using a parallel auxiliary line.

(b) In $\triangle ABC$, $\angle B = 2\angle A$ and $\angle C = 3\angle A$. Find all three angles and classify the triangle.

[4 Mark]

19. Write down the step-by-step procedure for constructing $\triangle ABC$ where side $AB = 6\text{ cm}$, $\angle A = 60^\circ$, and $\angle B = 45^\circ$ using a ruler, protractor, and compass. Calculate the measure of the third angle $\angle C$.

[4 Mark]

SECTION E: CASE-STUDY / COMPETENCY-BASED QUESTION (4 Marks)

20. **Case Study: Solar Panel Roof Framing**

A solar installation crew mounts photovoltaic panels on a rooftop triangular truss structure ABC :

- Base beam BC is horizontal.
- Support strut AB makes an inclination of 35° with the horizontal ($\angle B = 35^\circ$).
- Ridge angle at vertex A is designed to be 85° ($\angle A = 85^\circ$).

(a) Calculate the tilt angle of strut AC with the horizontal roofline ($\angle C$).

[1 Mark]

(b) If an exterior gusset plate at vertex B measures the exterior angle, what is its value?

[1 Mark]

(c) If strut $AB = 4.2\text{ m}$ and strut $AC = 2.8\text{ m}$, what is the theoretical maximum allowable length for the base beam BC to keep the truss rigid?

[2 Mark]